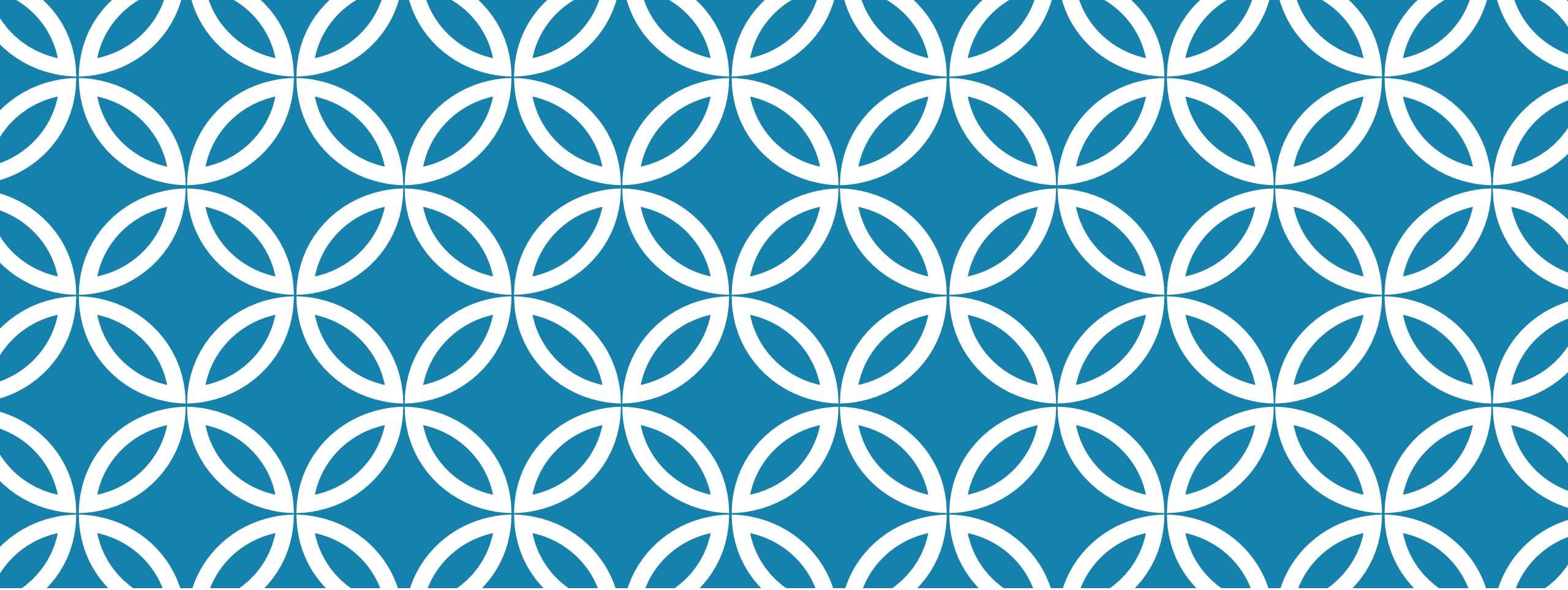
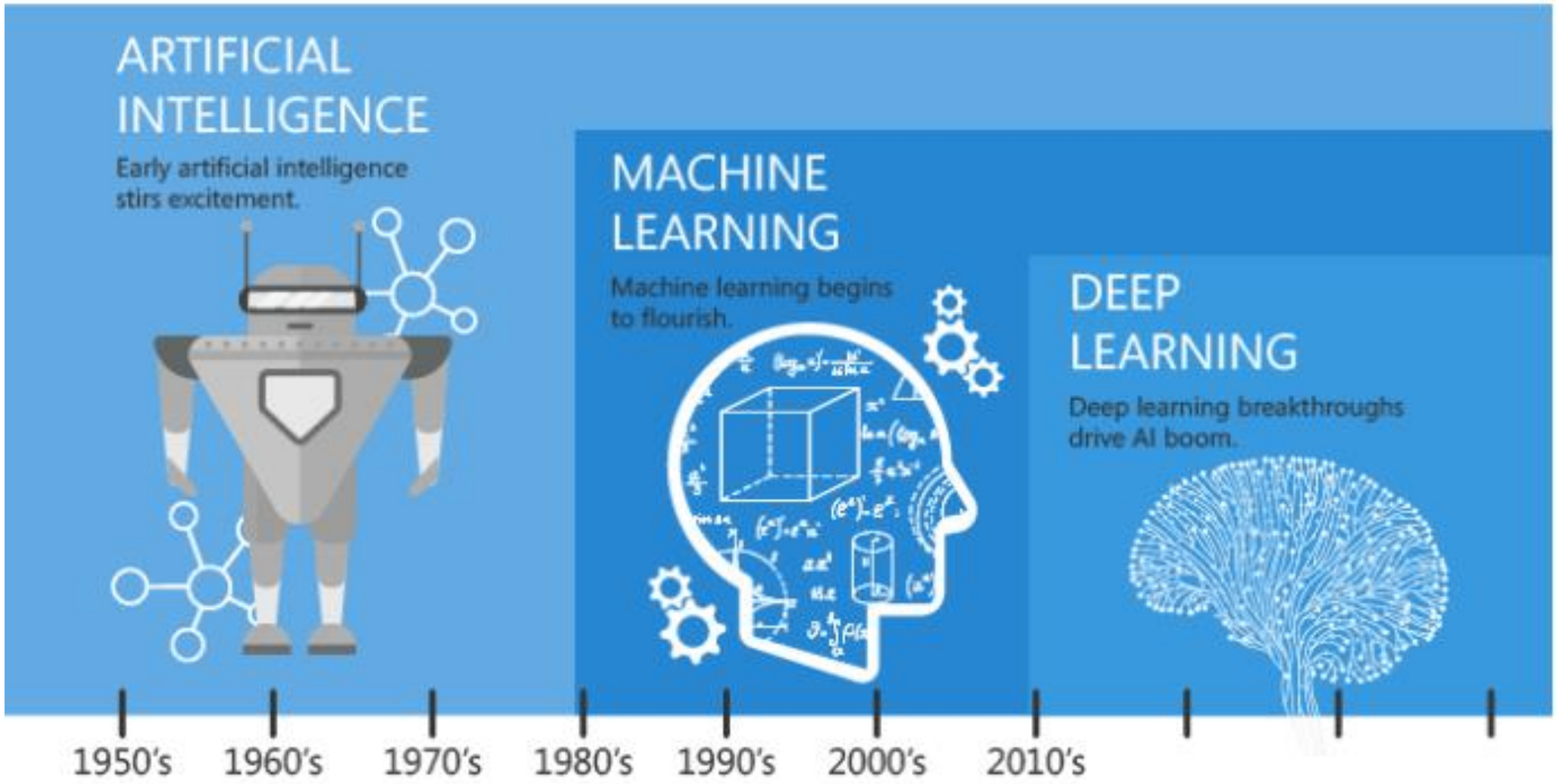


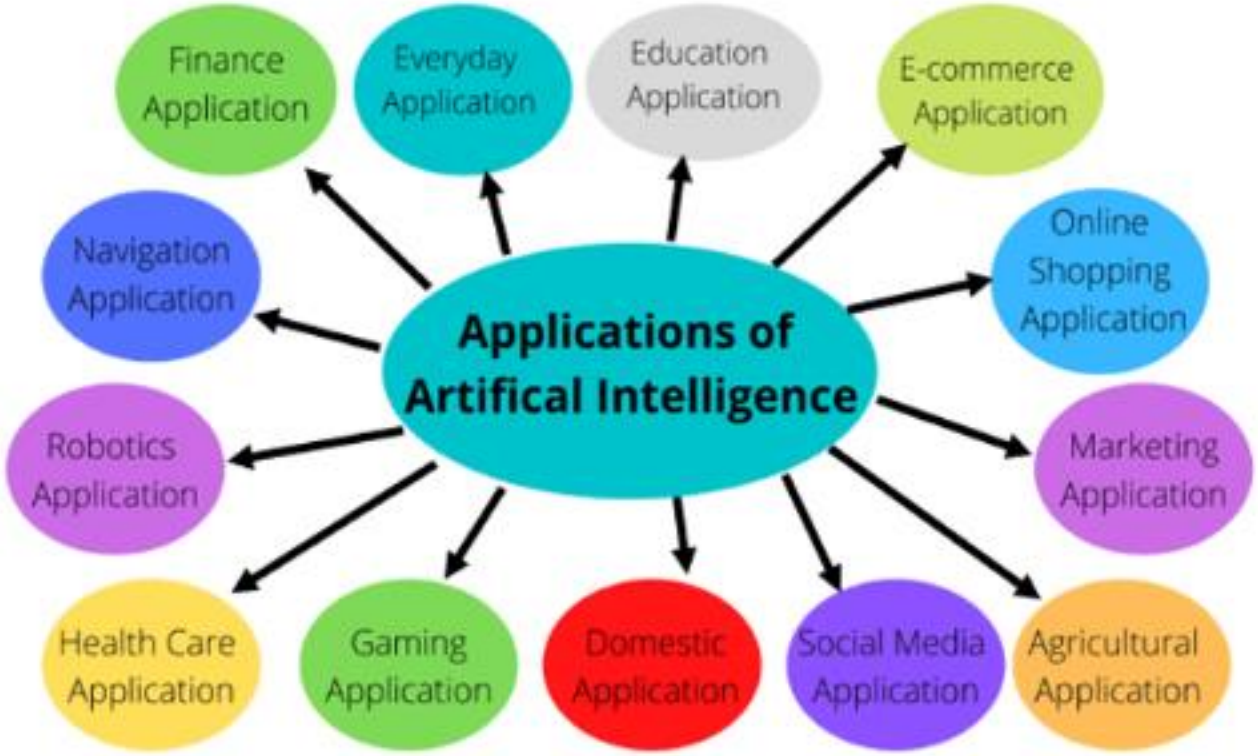
INTRODUCTION TO MACHINE LEARNING

Dr.Ameni Youssfi Nouira

HISTORY


ARTIFICIAL INTELLIGENCE
Early artificial intelligence stirs excitement.
MACHINE LEARNING
Machine learning begins to flourish.
DEEP LEARNING
Deep learning breakthroughs drive AI boom.
1950's
1960's
1970's
1980's
1990's
2000's
2010's

APPLICATIONS OF ARTIFICIAL INTELLIGENCE



MOTIVATIONS

- **Data is everywhere**

  Modern systems generate massive amounts of data in fields such as healthcare, finance, social networks, and industry. Machine Learning provides the tools to **extract knowledge and make intelligent decisions from data**, where traditional methods fail.

- **Explicit rules are no longer sufficient**

  Many real-world problems cannot be solved by manually defining rules:

  - image recognition
  - speech and text understanding
  - behavior prediction

➔ Machine Learning allows systems to **learn patterns automatically from examples.**

MOTIVATIONS

- **ML develops a powerful way of thinking**/ Machine Learning trains students to:

  - reason under uncertainty
  - work with imperfect models
  - evaluate performance rather than exact answers

➔This mindset is valuable far beyond ML itself.

MOTIVATIONS

- **ML connects theory with real-world applications**/ Machine Learning builds a bridge between:
  - Mathematics (linear algebra, probability, optimization)
  - Computer science (algorithms, data structures)
  - real-world problem solving

WHAT'S ML

« field of study that gives computers the ability to learn without being explicity programmed »

« the goal of ML is to build computer systems that can adapt and learn from their experience »

« A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E »

WHAT'S ML

- Automating automation
- Getting computers to program themselves
- Writing software is the bottleneck

  ➔ let the data do the work instead!

WHY ML IS IMPORTANT

Some tasks cannot be defined well, except by examples (recognizing people)

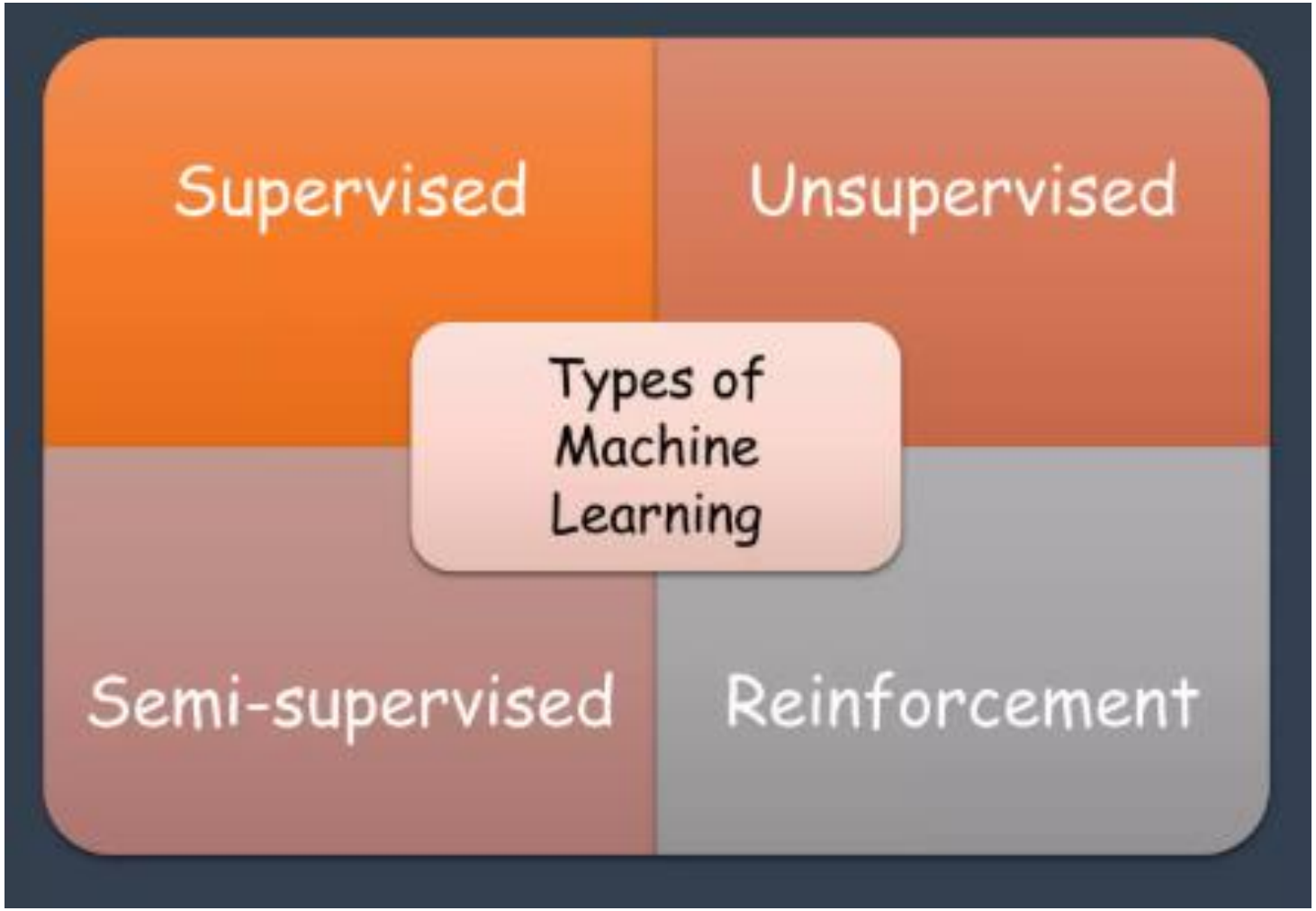
Relationships and correlations can be hidden within large amounts of data, ML/datamining may be able to find these relationships

Human designers often produce machines that do not work as well as desired in the environments in which they are used,

GENERALIZATION

A core objective of learner is to generalize from its experience, Generalization in this context is the ability of a learning machine to performm accurately on new, unseen examples/tasks after having experienced a learning data set.

ML ALGORITHMS TYPES



SUPERVISED LEARNING

Algorithms are trained on labeled examples i,e; input where the desired output is known,

In supervised learning, you train the machine using data which is well « labeled »

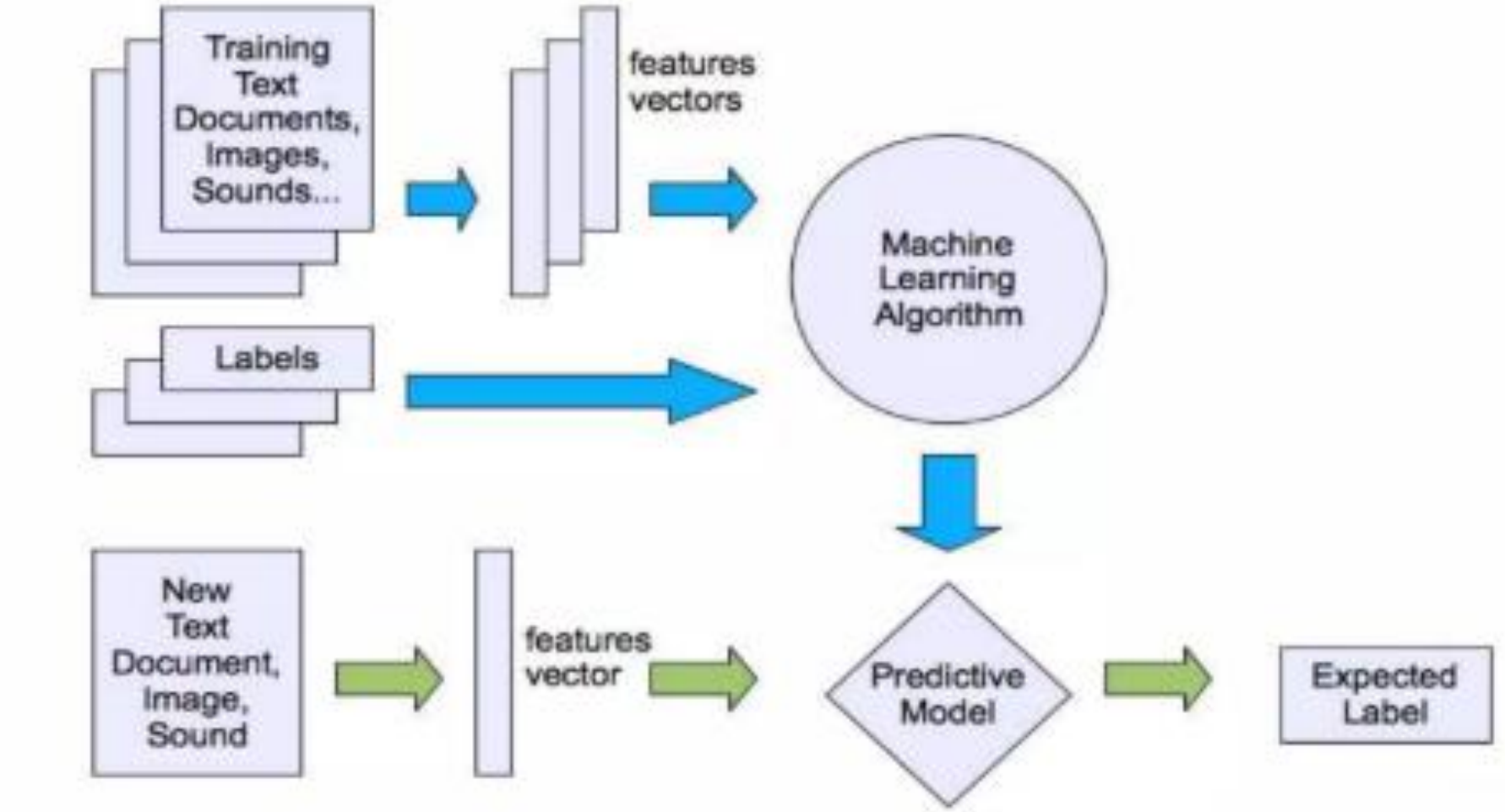
It means data is already tagged with the correct answer.

It can be compared to learning which takes place in the presence of a supervisor or a teacher

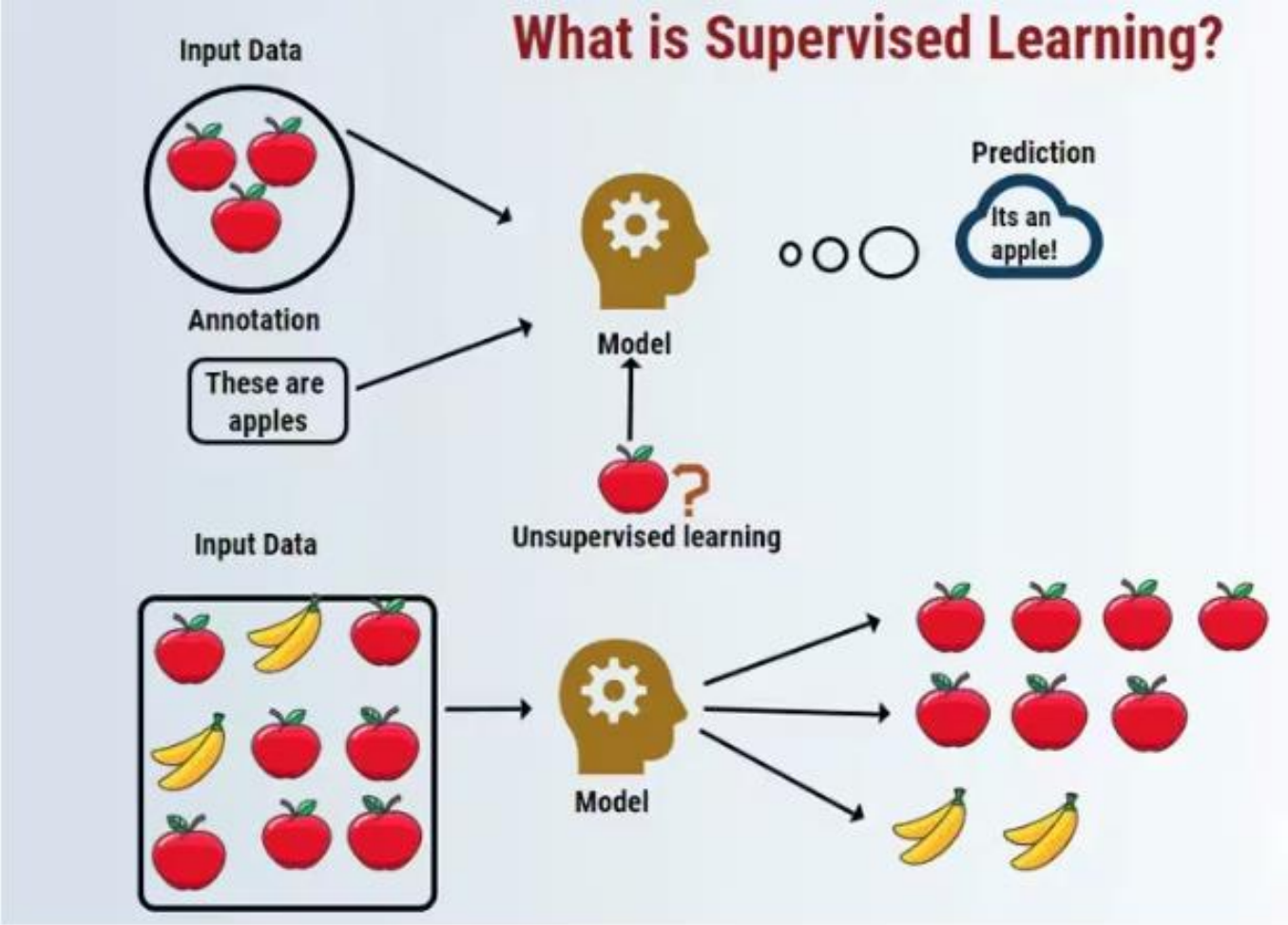
A supervised learning algorithm learns from labeled training data, helps you to predict outcomes for unforseen data

One disadvantage of this learning method is that the dataset has to be hand-labeled either by a machine learning Engineer or a data scientist this is a very costly process, especially when dealing with large volumes of data

SUPERVISED LEARNING



SUPERVISED LEARNING



UNSUPERVISED LEARNING

Algorithms operate on unlabelled examples, ie; input where the desired output is unknown

Unsupervised Larning is a machine learning technique in which the users do not need to supervise the model,

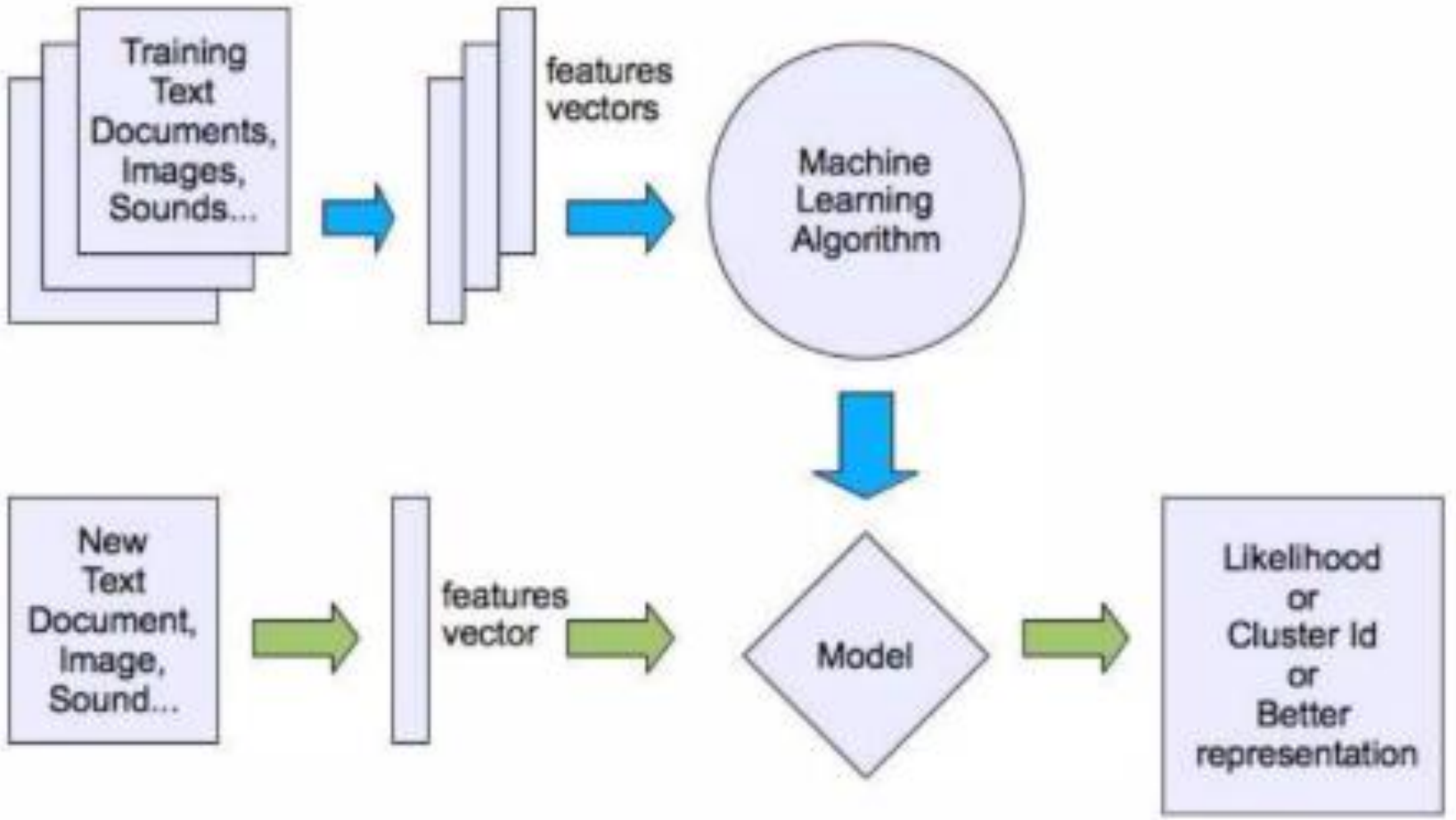
Instead, it allows the model to work on its own to discover patterns and information that was previously undetected

It mainly deals with the unlabeled data

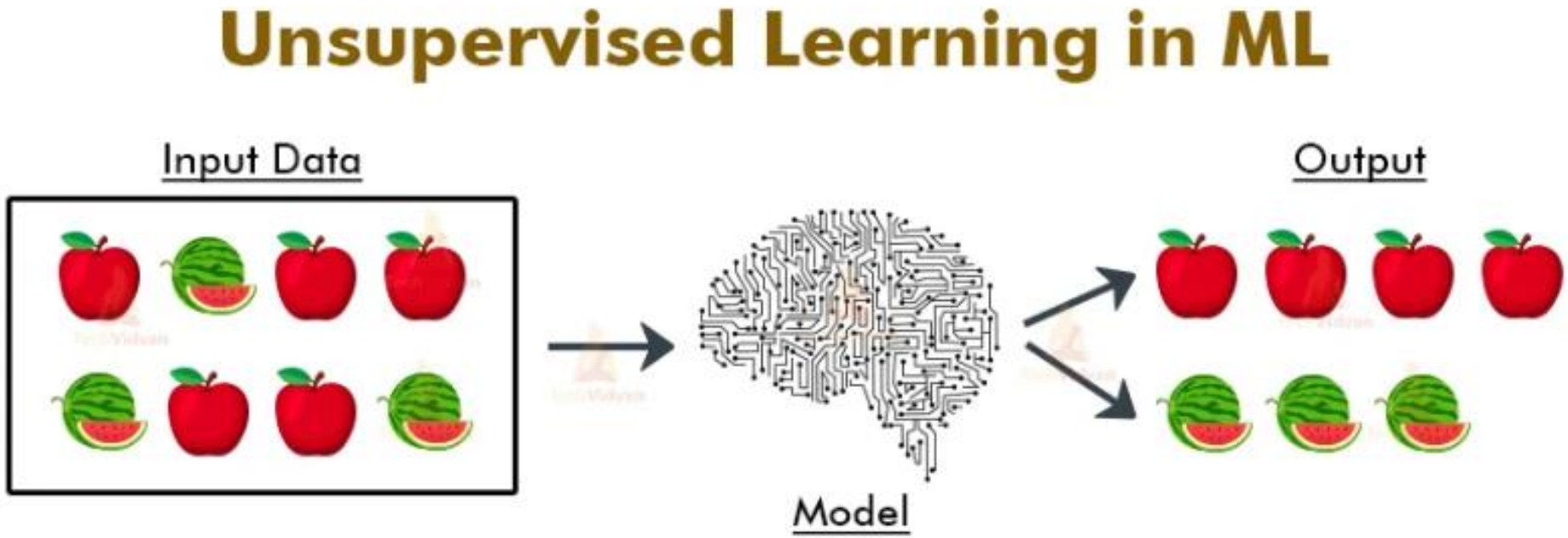
Unsupervised learning problems are grouped into clustering and association problems

The most basic diadvantage of any unsupervised learning is that it's application spectrum is limited

UNSUPERVISED LEARNING



UNSUPERVISED LEARNING



SEMI-SUPERVISED LEARNING

Semi-supervised learning combines both labeled and unlabelled examples to generate an appropriate function or classifier,

SEMI-SUPERVISED LEARNING is the type of ML that uses a combination of a small amount of labeled data and a large amount of unlabeled data to train models,

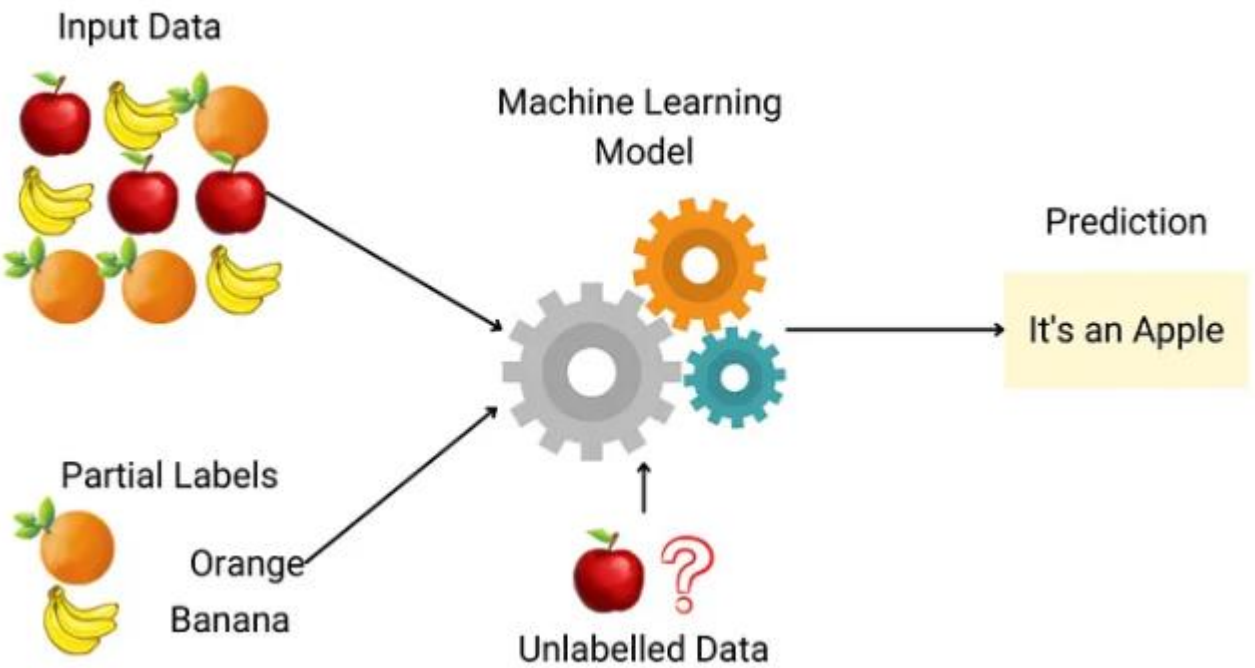
Intuitively, one may imagine the three types of learning algorithms as:

Supervised learning where a student is under the supervision learning of a teacher at both home and school,

Unsupervised learning where a student has to figure out a concept himself

Semi-supervised learing where a teacher teaches a few concepts in class and gives questions as homework which are based on similar concepts

SEMI-SUPERVISED LEARNING



REINFORCEMENT LEARNING

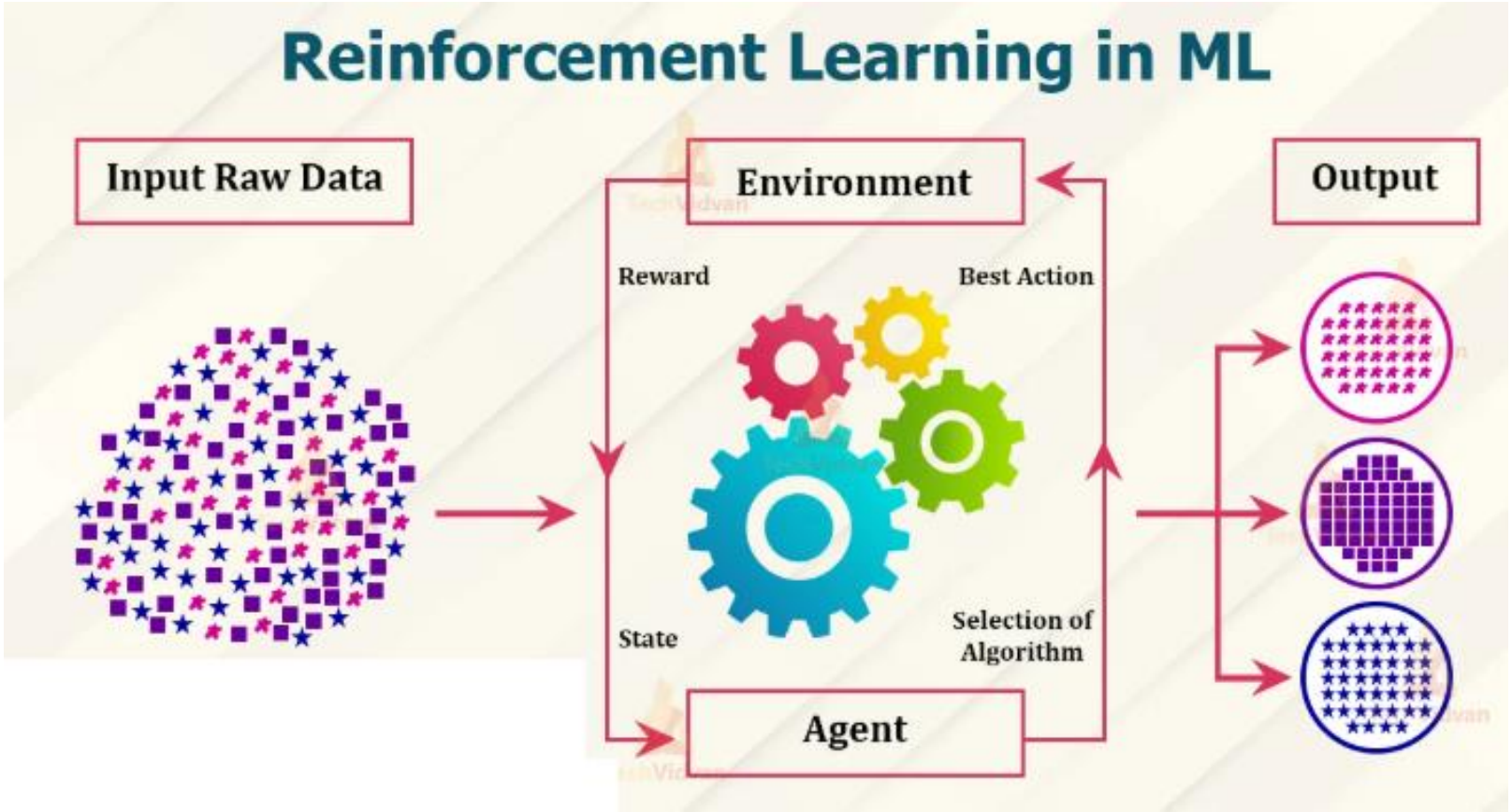
RL is a feedback-based learning method, in which a learning agent gets a reward for each right action and gets a penalty for each wrong action,

The agent learns automatically with these feedbacks and improves its performance,

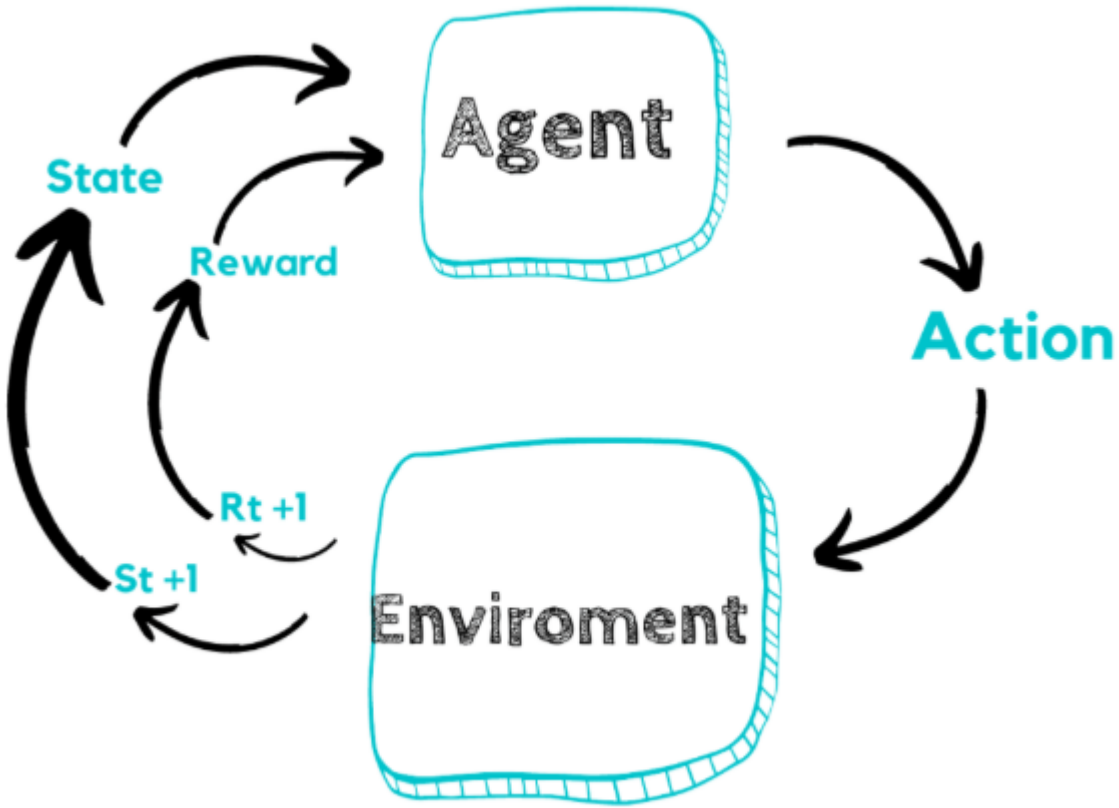
In RL the agent interacts with the environment and explores it,

The goal of an agent is to get the most reward points, and hence, it improves its performance,

RL


Reinforcement Learning in ML
Input Raw Data
Environment
Output
Reward
Best Action
State
Selection of Algorithm
Agent

RL



DATA

In today's digital world, we are surrounded by data everywhere, there is a huge amount of data being generated everyday,

In simple words, any information about anything can be termed DATA

MI algorithms are constantly analyzing & learning from this data to improve their future predictions and outcomes automatically

We can say, data is at the heart of the ML and it is the data which enables ML to do all amazing things it does

CONTD...

Why did Facebook acquire Whatsapp by paying a huge price of $19 billion?

The answer is very simple and logical, it is to have access to the user's information that Facebook may not have bu whatsApp will have, This information of their users is of paramount importance to Facebook as it will facilitate the task of improvement in their services



CONTD...

Data: Data is a collection of information about something,for example, the features of smart phone: screen size, memory, processor, brand namen, and its color

Information: data that has been interpreted and manipulated and has now some meaningful inference for the users,

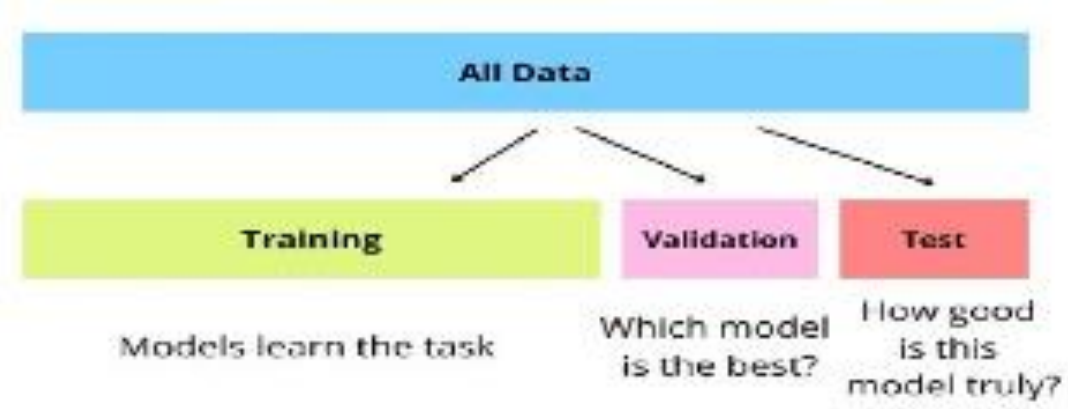
Knowledge: Combination of inferred information, experiences, learning, and insights, results in awareness or concept building for an individual or organization

HOW TO SPLIT DATA IN ML

Training_Data: the part of the data that we use to train our model, this is the data that your model actually sees and learns from,

Validation Data: Dara that is used to do a frequent evaluation of the model, this data plays its part when the model is actually training,

Testing-data: Once our model is completely trained, testing data provides and unbiased evaluation,



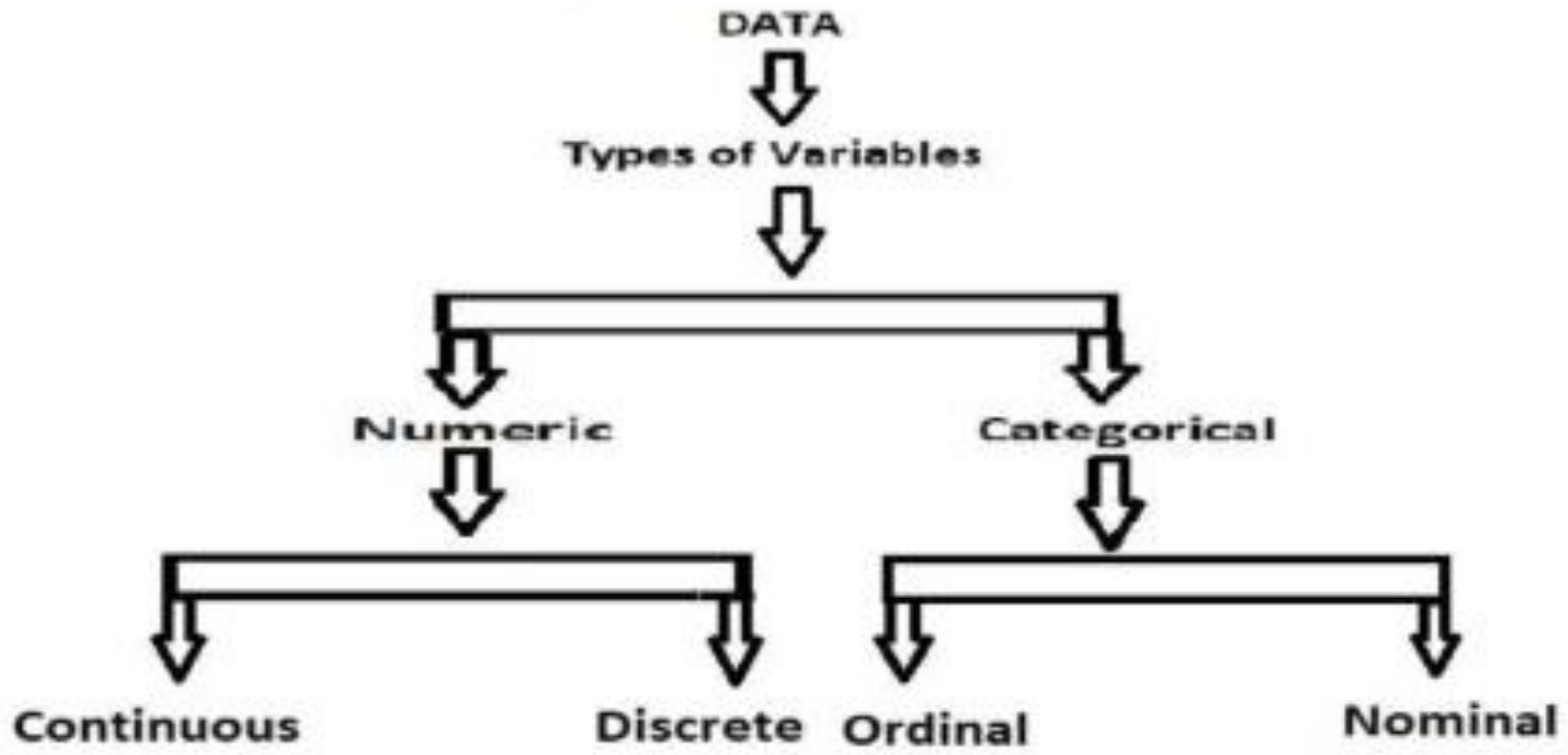
VARIABLE

Variable

- A variable represents one specific characteristic of the data and tells us specific information about the data under consideration.
- Variables are also known as Data items, as they constitute the data,
- A variable can take any value like numbers or characters,
- A variable can take different values depending on the observation. For example, variables such as **age, height, weight,** or **salary** can have different values for different individuals.
- It is called a variable because its value can change across different data points or over time

There are two types of variables

DIFFIRENT FORMS OF DATA



CONTD...

1/ Numeric or Quantitative variable

One which can be measured or counted is known as Numeric or Quantitative variable

For example, exact price of phone in rupees

Numeric variable are subdivided into two categories

A) continous variables:

Continous Variables: are those numeric variables which can take any value between a certain set of real numbers i,e; integers, fractions, decimals and so on,

Their values will remain in that finite interval

For example, height, temperature, time, age….

CONTD...

B) Discrete Variables

Discrete variables are those which are countable and can take only whole numbers (integers) as value

For example, number of contacts on phone, number of calls made in a day etc are all whole numbers and each of them are distinct for each scenario,

CONTD

Student ID	Name	Gender (Nominal)	Grade Level (Ordinal)	Number of Courses (Discrete)	GPA (Continuous)
1	Ali	Male	Beginner	3	2.8
2	Sara	Female	Intermediate	5	3.4
3	Omar	Male	Advanced	4	3.9
4	Lina	Female	Beginner	2	2.5
5	Youssef	Male	Intermediate	6	3.2

CONTD

Nominal: Gender (Male, Female) → no order

Ordinal: Grade Level (Beginner < Intermediate < Advanced) → has order

Discrete: Number of Courses (2, 3, 4…) → countable values

Continuous: GPA (2.5, 3.4, 3.9…) → decimal values

CONTD...

2. ***Categorical or Qualitative Variable***

One which represents a characteristics and cannot be measured or counted is known as Categorical or Qualitative variable

Categorical variable are those which are like adjectives and express a feeling or characteistics, but they cannot be quantified or measured,

For example, cheap mobile phone is categorical variable because it does not define exact price

Categorical variables are sub divided nto two categories

CONTD...

A) Ordinal Variable

Ordinal variables are those categorical variables which can be arranged in a logical order i,e; they can be ranked in ascending or descending order based on their values,

For example, product rating Poor>Average>Good

B) Nominal Variables:

Nominal Variables are those categorical variables which cannot be logically ranked based on their values

For example, colour of your phone (Black, white,,,,)

OTHER FORMS OF DATA

A) Labelled Data:

The data which contains a target variable or an output variable that answers a question of interest is called a labelled data,

For example, results of a patient suffering from a particular disease and receiving treatment for it

B) Unlabelled Data:

Unlabelled data is one which contains information about something but does not have a predefined target variable

It is actually just opposite of labelled data

For example, population data of a particular state

OTHER FORMS OF DATA

Semi-Labeled Data is a dataset where **only some of the data is labeled,** while the rest is unlabeled.

Weakly Labeled Data is data that has **labels, but they are not very accurate or precise.**

Noisy Data that contains **errors, incorrect values, or irrelevant information that can affect model performance.**

PROPRETIES OF DATA

1. Volume : Scale of Data, with the growing world population and technology at exposure, huge data is being generated each and every millisecond
2. Variety: Different forms of data /healthcare, image videos, audio clipping.
3. Velocity: Rate of data streaming and generation
4. Value : Meaningfulness of data in terms of information that researchers can infer from it.
5. Veracity: certainty and correctness in data we are working on.

CONCLUSION

Machines should be able to do all things what we can do & machine learning will play a big role in achieving this goal.